

M2 Mock 1 : JOIN3_R_2: Products With Total Quantity Sold And Customer Names

Database Description

The retail_store database is structured to manage and analyze key transactional and operational data within a retail environment. It captures detailed information about customers, product inventory, orders, and item-level purchases. This schema supports querying customer purchase behavior, product sales performance, and order histories, enabling insights into customer trends, product demand, and overall business operations.

Schema Specification

Table: Customers

Stores customer profile and contact information.

Column Name	Data Type	Constraints	Description
customer_id	INT	PRIMARY KEY	Unique ID for each customer
name	VARCHAR(100)	NOT NULL	Customer full name
email	VARCHAR(100)	UNIQUE NOT NULL	Email address of the customer
phone	VARCHAR(20)		Contact number

Table: Products

Stores product catalog and pricing information.

Column Name	Data Type	Constraints	Description
product_id	INT	PRIMARY KEY	Unique product identifier
name	VARCHAR(100)	NOT NULL	Name of the product
category	VARCHAR(50)		Category to which product belongs
price	DECIMAL(10,2)	NOT NULL	Price per unit

Table: Orders

Captures order-level metadata including customer and date.

Column Name	Data Type	Constraints	Description
order_id	INT	PRIMARY KEY	Unique order number
customer_id	INT	FOREIGN KEY → Customers(customer_id)	Customer who placed the order
order_date	DATE	NOT NULL	Date the order was placed
total_amount	DECIMAL(10,2)		Total bill amount for the order

Table: Order_Items

Represents items associated with each order, along with quantity and pricing.

Column Name	Data Type	Constraints	Description
order_id	INT	FOREIGN KEY → Orders(order_id)	Order to which item belongs
product_id	INT	FOREIGN KEY → Products(product_id)	Product purchased in the order
quantity	INT	NOT NULL	Number of units ordered
unit_price	DECIMAL(10,2)	NOT NULL	Price per unit at the time of order
PRIMARY KEY	Composite	PRIMARY KEY (order_id, product_id)	Ensures one entry per product/order

✓ Test Data

Customers

customer_id	name	email	phone
201	John Doe	john@example.com	1234567890
202	Jane Smith	jane@example.com	2345678901
203	Emily Davis	emily@example.com	3456789012
204	Mark Wilson	mark@example.com	4567890123

Products

product_id	name	category	price
301	Laptop	Electronics	1000.00
302	Smartphone	Electronics	600.00
303	Office Chair	Furniture	150.00
304	Notebook Set	Stationery	10.00

Orders

order_id	customer_id	order_date	total_amount
401	201	2023-08-01	1010.00
402	202	2023-08-03	1600.00
403	203	2023-08-04	20.00
404	204	2023-08-05	750.00

Order_Items

order_id	product_id	quantity	unit_price
401	301	1	1000.00
401	304	1	10.00
402	301	1	1000.00
402	302	1	600.00
403	304	2	10.00
404	303	5	150.00
404	304	1	10.00
402	303	1	150.00
403	302	1	600.00
401	303	2	150.00
403	301	1	1000.00
404	302	1	600.00

Problem Statement

Management wants a report that shows each product, the **total quantity sold**, and the **names of customers who purchased it**. This will help in analyzing product popularity and customer purchase behavior.

Query to be Implemented

Write a SQL query to retrieve:

- product_name
- total_quantity_sold (sum of all quantities sold)
- customer_name

Each row should represent a product and one of its purchasing customers.

Expected Output

product_name	total_quantity_sold	customer_name
Laptop	3	John Doe
Laptop	3	Jane Smith
Laptop	3	Emily Davis
Notebook Set	4	John Doe
Notebook Set	4	Emily Davis
Notebook Set	4	Mark Wilson
Office Chair	8	John Doe
Office Chair	8	Jane Smith
Office Chair	8	Mark Wilson
Smartphone	3	Jane Smith
Smartphone	3	Emily Davis
Smartphone	3	Mark Wilson